



Bayer's *in silico* ADMET platform: a journey of machine learning over the past two decades

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Over the past two decades, an *in silico* absorption, distribution, metabolism, and excretion (ADMET) platform has been created at Bayer Pharma with the goal to generate models for a variety of pharmacokinetic and physicochemical endpoints in early drug discovery. These tools are accessible to all scientists within the company and can be a useful in assisting with the selection and design of novel leads, as well as the process of lead optimization. Here, we discuss the development of machine-learning (ML) approaches with special emphasis on data, descriptors, and algorithms. We show that high company internal data quality and tailored descriptors, as well as a thorough understanding of the experimental endpoints, are essential to the utility of our models. We discuss the recent impact of deep neural networks and show selected application examples.

Introduction

ADMET are crucial parameters for the discovery and optimization of new drugs. This has long been recognized, and pharmaceutical companies have invested heavily to develop new assays and to ramp up their testing capacity, enabling them to characterize thousands of compounds in high-quality *in vitro* ADMET assays. The stored structure–activity/structure–property relationship (SAR/SPR) data can be a treasure and have the potential to impact research beyond the specific project for which these measurements were conducted. Computational groups have since been using those data to generate an understanding for the principles underlying certain ADMET endpoints and to develop *in silico* models that can serve as an additional tool to assist researchers in their quest for new compounds [1–5]. The primary goal of those models is not to reduce the overall number of *in vitro* or *in vivo* ADMET

experiments, but to allow scientists to better focus their experiments on the most promising compounds.

In this review, we discuss the development of *in silico* ADMET approaches generated at Bayer over the past 20 years. There are two conceptually different approaches to *in silico* ADMET. The first is a protein structure-based approach, in which the interaction of compounds with defined proteins, important for ADMET properties, is modeled and used to design better compounds. It requires ADMET effects clearly associated with single ADMET-relevant proteins (e.g., cytochrome P450 enzymes, PXR, hERG, PgP, or HSA) and high-resolution X-ray structures of those proteins. The application and utility of such approaches has been discussed elsewhere [6,7]. Here, we concentrate on the second conceptual approach, in which measured *in vitro/in vivo* data of many compounds are used to build models using ML (Fig. 1). We summarize the experience at Bayer over 20 years in ML by pointing out our ‘working-horse’ approaches and we discuss the latest developments in tailored descriptors and algorithms, such as deep

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In case the assay protocol changes, different scenarios can be applied. Depending on the assay, the data are merged in such a way that the values stay comparable, if possible. This always requires close collaboration with experimentalists to gain a deeper understanding of assay setup and interpretation of the results. This is more challenging if the data are used for regression rather than for classification models.

If it is not possible to merge the data in a meaningful way, there is the option to use both assays as different tasks in a multitask model. Only as worst case, we would neglect the old data coming from a previous assay setup and create a completely new model.

The importance of data curation and extension is nicely illustrated by two examples. In a collaboration with Simulations Plus, we provided ~19 500 compounds with experimental pK_a values and, by this, reduced the mean absolute error from 0.72 to 0.5 log units [15], resulting in one of the best-performing pK_a models [16]. The rigorous curation of the Accelrys Metabolite Database yielding 18 000 high-quality metabolic reactions provided a quality increase in the Site-of-Metabolite (SoM) models [17] (see 'Applications' section) compared with the former CypScore [18] based on only 2400 transformations.

Generally, the larger the data set, the more diverse the chemistry, the broader and the more evenly the distribution of the values; in addition, the lower the experimental error, the better

performing the model. Over the years, with an increasing size of our proprietary data set, we were able to turn models from classifiers into regression models (Fig. 2).

Descriptors

Molecules are dynamic multiconformational 3D entities comprising nuclei and electrons. Translation into machine-readable descriptors will result in information loss. Descriptors can be classified by increasing complexity into 1D constitutional descriptors, such as molecular weight, 2D or topological descriptors (e.g., fingerprints), and 3D descriptors, based on molecular fields or quantum chemistry from 3D structures [19].

Our work-horse descriptors since 2001, confirmed by many publications, are circular extended connectivity [20] fingerprints (ECFP), which encode properties of atoms and their neighbors into a bit vector of certain radius, feature type, and fold. The best setting for each task, depending on structure diversity and endpoint, always has to be empirically identified by trial.

Nevertheless, we asked ourselves: can we do better? Inspired by machine translation models [21], we developed a method to encode the SMILES of a molecule into a 512D continuous space, and a recurrent network to translate that embedding back into a canonical form of SMILES [22]. This type of network, depending only on chemical structure, can be trained on very large data

		Insufficient quality	First approach	Medium model	Good model	Robust model		
Endpoint		Model type	Data set size	2005	2009	2014	2019	Retraining
Absorption	Caco-2 permeation	C (N)	>10 000			RF	SVR	Weekly
	Caco-2 efflux	C (N)	>10 000			RF	SVR	Weekly
	Bioavailability (rat)	C	~2000				RF	On demand
Distribution	Human serum albumin	N	>30 000			PLS	MTNN	On demand
	Fraction unbound	N	>1000			PLS	MTNN	On demand
Metabolism	Microsomal stability (hum)	C (N)	>10 000			RF	RF	Weekly
	Microsomal stability (mouse)	C (N)	>10 000			RF	RF	Weekly
	Microsomal stability (rat)	C (N)	>10 000			RF	RF	Weekly
	Hepatocyte stability (rat)	C (N)	>30 000			RF	RF	Weekly
Toxicity	hERG inhibition	C	>10 000			RF	SVM	Weekly
	Ames mutagenicity	C	>10 000			RF	RF	On demand
	CYP inhibition isoforms	C	>10 000			RF	RF	On demand
	Phospholipidosis	C	<1000			SVM	SVM	On demand
	Structure filter tool	Score	n.a.	-	-	-	-	On demand
PhysChem	Solubility (DMSO)	N	>30 ,000			PLS	MTNN	On demand
	Solubility (Powder)	N	<10 000				MTNN	On demand
	logD @ pH 7.5	N	>70 000			PLS	MTNN	On demand
	Membrane affinity	N	<10 000			PLS	MTNN	On demand
	pKa	N	>10 000			ANN	ANN	On demand
	Oral PhysChem score	Score	n.a.	-	-	-	-	On demand
	i.v. PhysChem score	Score	n.a.	-	-	-	-	On demand

Drug Discovery Today

Drug Discovery Today

FIGURE 2

Bayer *in silico* absorption, distribution, metabolism, and excretion (ADMET)/physicochemical model portfolio. Endpoint describes the respective assay, model type C (Classifier), N (Numerical); C(N) is a numerical model presented to the user as classifier; Score as described [50,63]. Algorithm: PLS partial least squares, RF random forest, SVR support vector regression, SVM support vector machine, ANN artificial neural network, MTNN multi-task neural network. Generally, MTNN refers to a multi-task model learning the properties logD, melting point, solubility, serum albumin binding, and membrane affinity together. Fraction of unbound of human, rat, mouse, and monkey is learned together with logD as a distinct MTNN. Historical development of model quality from 2005 onwards is shown via color code from red (insufficient quality) light green (good model: high-quality classifier and medium-quality numeric models) to dark green (robust model: high-quality numeric models). These scores reflect the relative robustness of the models for the different endpoints. The low quality of the mouse microsomal stability models is speculated to be linked to a lower quality of these data compared with other species.

sets of tens of millions of structures. The resulting molecular descriptor was useful for building quantitative structure–activity relationship (QSAR) models (especially in combination with support vector machines) and for virtual screening [23]. The presence of the decoder that can map the embedding of a compound to its chemical structure, is also significant in the sense that the chemical representation is now reversible. This means that navigating the embedding space in directions that improve the properties of interest can lead to the generation of new ideas for lead optimization projects, after decoding the optimized embedding [22].

Alternatively, a molecule can be seen as a graph, with atoms being the vertices and bonds the edges. This opens another way to generate machine-learned descriptors. Graph convolutional networks [24] are a specific neural network architecture that can learn vertex (and bond) features in an end-to-end fashion. The feature representation of each node is aggregated by summing or averaging with those of its neighbors using a so-called ‘adjacency matrix’. The aggregated nodes are then fed into neural networks applying affine transformation with learned weights followed by a nonlinear activation function. The effect is that the atom features are learned and the neighboring atoms in the graph can influence each other. This approach has been implemented specifically for chemistry applications [25–27]. However, because the training is end-to-end (meaning the features are tailored to the problem at hand), large training sets are needed to avoid overfitting.

Earlier, we discussed molecular descriptors (i.e., describing the molecule as a whole). By contrast, SoM modeling of the regioselectivity of reactions catalyzed by metabolic enzymes requires ML on individual atoms, not whole molecules. We recently developed anisotropic atomic reactivity descriptors from quantum mechanical atomic charges [28]. A SoM model was trained using ~18 000 chemical transformations of 18 different enzymatic reactions, said descriptors, and random forests [17,29]. The model performs well with a precision of 0.51 for a hard test set extracted from the most recent literature (see ‘Application Examples’ section). Over the past year, the descriptor was further applied in an attempt to assess the risk of Ames mutagenicity of primary aromatic amines [30] and the strengths of hydrogen bonding [31–33].

Especially with regard to compounds beyond the rule-of-five [34,35] chemical space, and the recent interest in macrocycles, it is believed that descriptors derived from 2D representations are not sufficient. Starting from 3D structures adds one level of complexity because of conformational ensembles [36], but enables researchers to derive descriptors for the fluctuation of, for example, the polar surface area [37] or (intramolecular) hydrogen bonding [17,29,33,38]. One such descriptor is the MDPF derived from molecular dynamics for modeling solvation-free energies and distribution coefficients [39]. Clever descriptions of 3D features of molecules would constitute one approach towards the improvement of *in silico* ADMET and other ML models.

Algorithms

The dependence between descriptors and endpoints is nonlinear, requiring also nonlinear algorithms. Our observation over the years [40], again supported by many publications, is that support vector machines [41] and random forests [42] are typically among

the most useful algorithms. Partial least squares [43] sometimes yields models that are more stable over longer periods of time. Regression models are always preferred over classifiers and, therefore, new algorithms are constantly evaluated.

Over the past 5 years, there has been a steep increase in the use of deep neural networks in computational chemistry [44]. They are well suited for multitask learning and are promising because they can extract chemical features unbiased by the choice of particular fingerprints (see ‘Descriptors’ section). An intriguing example is the modeling of cellular toxicity [45] or the Bayer model for rat oral bioavailability from chemical structures [46]. Whereas the classical approach of fingerprint and/or random forest was identical in classification performance to deep learning, regression models for exposure were only achievable with a deep-learning approach.

Deep learning also makes multitask learning (i.e., simultaneously learning several related tasks in one model) very natural. We performed comparisons on in-house ADMET data sets and found that, in general, in line with earlier publications [47,48], deep neural networks outperform random forest or similar methods by a small margin, whereas combined training of related endpoints allows for tasks with fewer measurements to benefit from the joined learned representations in a multitask setting [49]. Particularly for physicochemical properties, combining graph convolutional networks with multitask training led to a significant increase in performance for all modeled endpoints compared with the original methodology [24]. One of the endpoints that especially benefitted from this new multitask deep-learning network was solubility. Here, we were able to replace the former classification with a regression model for solubility (Fig. 2).

The need for novel methods that allow for good regression models is based on the requirement to feed the model outcomes into multiple parameter optimization processes, which does not work with classifiers. Numerical data also allow for more flexible decision making because each regression model can be turned into a classification model at any desired threshold.

For endpoints with few data points, imbalanced, or censored data, for which it is challenging to build good regression models, we can still derive robust classifications. Other candidates for classification models are endpoints with a binary output, such as phospholipidosis (Fig. 2). Finally, traffic lights [50] are often the clearest and most user-friendly visualization, especially if they are validated with meaningful druglike datasets.

Quality assessment of models

Proper evaluation of models via nested cross-validation (CV) and on independent test sets is crucial to ensure robust modeling outside the chemical space used for training. Different metrics have to be applied for categorical or regression problems.

Common metrics for classification models are derived from the so-called ‘confusion matrix’, which provides numbers for true positives, true negatives, false positives, and false negatives and, as derivatives, overall accuracy, sensitivity, specificity, negative/positive precision values, and the Matthews Correlation Coefficient (MCC). Another popular metric is the area under the ROC curve (AUC), which gives the classification performance of the model for every possible class threshold, including the parameters true positive and false positive rate. For regression models, common metrics are R^2 (coefficient of determination R-squared),

root mean square error (standard deviation of residuals), and Spearman's rho (nonparametric rank correlation coefficient).

Besides choosing the right metrics, also well-chosen statistical validation techniques are crucial to find a suitable balance between the two extremes of overfitted and underfitted models [51]. We typically put 20% of the data aside as external test set for the quality of the final model. The other 80% is used as training data in a CV set up. CV with random split is inadequate for pharma-like chemical congeneric series. Chronological 'time-dependent' CV or 'leave-cluster-out' CCV are more realistic estimators for the ability of a model to extrapolate into new unforeseen chemical spaces. K-Means clustering is our preferred method for the 'leave-cluster-out' validation because it is convenient to compute and leads to good results, but any clustering method can be used. Which type of CV shows the most realistic predictivity is data set dependent. Whereas our best logD (pH7.4) model is relatively robust with a leave-one-cluster-out CV R^2 of 0.88 compared with 0.91 at random split [49], our at all times best DMSO solubility model showed R^2 of 0.59 in leave-one-cluster-out but only 0.68 at random split CV.

Our current minimum quality requirements for implementation are primarily based on leave-one-cluster-out CV. For classification, the MCC should be >0.4 and for regression models the Pearson R^2 should be >0.3 and Spearman $R^2 >0.6$.

Rodgers *et al.* [52] reported the importance of regular model updates to keep them useful for current compounds. The achievable improvement from regular retraining depends on the model building technique used and on the studied property.

We have already implemented a weekly automated data download and filtering process and an automated model retraining for endpoints such as microsomal and hepatocyte stability (Fig. 2) and are currently implementing it for further endpoints. It is vital here

to include the time stamps of measured data to track retraining effects and check the quality of the automated models.

For instance, for metabolic stability, we observed that, over time, although global CV performance stays steady, retraining has a positive effect on modeling accuracy of upcoming chemical classes, even if a low number of molecules (<50) are being added to the training data. In our experience, the risk of frequent 'flipping' between classes, which might be annoying for the medicinal chemists, is rather low.

As mentioned earlier, the accuracy of property modeling for completely novel molecules has limitations because the drug-like [53] space is immense and our models are based on limited sets from our confined chemical space.

The test set-derived prediction error provides information on the average performance on this set, but not on the prediction reliability for individual future molecules. Therefore, many different so-called 'applicability domain' (AD) measures have been introduced in recent years, which can be grouped into two classes. Methods that apply distance measures on how well the future object is embedded in the training set, are termed 'novelty detection'. Methods that quantify the distance to the decision boundary of the classifier are called 'confidence estimation'. Whereas the former can be applied for any algorithm using, for example, cosine, Tanimoto, or Mahalanobis distance [54] to the full training set [55], the latter measures are algorithm dependent. Recently, it was shown that confidence measures are superior in general [56,57]. Alternative concepts, such as conformal prediction [58] or using one ML approach for estimation of the AD of another one [59], still have to prove their usefulness.

Most of the Bayer models provide algorithm-inherent reliability estimates together with the actual values. In case of random



FIGURE 3

Bayer models are made available to researchers via an in-house developed data informatics platform, a thick-client workbook with capabilities for data retrieval from the data warehouse, data visualization, compute tools, and document and project management interface. A typical visualization with spreadsheet and scatterplot is shown.

forests, this is the percentage of tree votes or, in the case of SVM, distance from the hyperplane. Predictions below certain predefined thresholds (e.g., 0.6 for random forest) are not reported. Multitask models do not yet provide a reliability metric.

Application examples

Our internal data informatics platform has become a useful tool for assisting the processes of lead selection, compound design, and synthesis planning. It allows all scientists at Bayer quick access to the newest models. A typical decision spreadsheet is shown in Fig. 3.

In the early hit-to-lead phase, hit lists are clustered and prioritized by Bayer expert teams using standardized reports that incorporate both experimental data and a full *in silico* profile, allowing documented data-driven decision taking. Our models are applied for *in silico* property characterization and help to assess virtual compounds during lead identification and optimization in a variety of drug discovery projects at Bayer (e.g., [60–62]). For example, during the refinement of PTGES inhibitors, experimental and predicted Caco-2 data and calculated pKa values for a series of 30 compounds implicated a narrow balance between acidity and

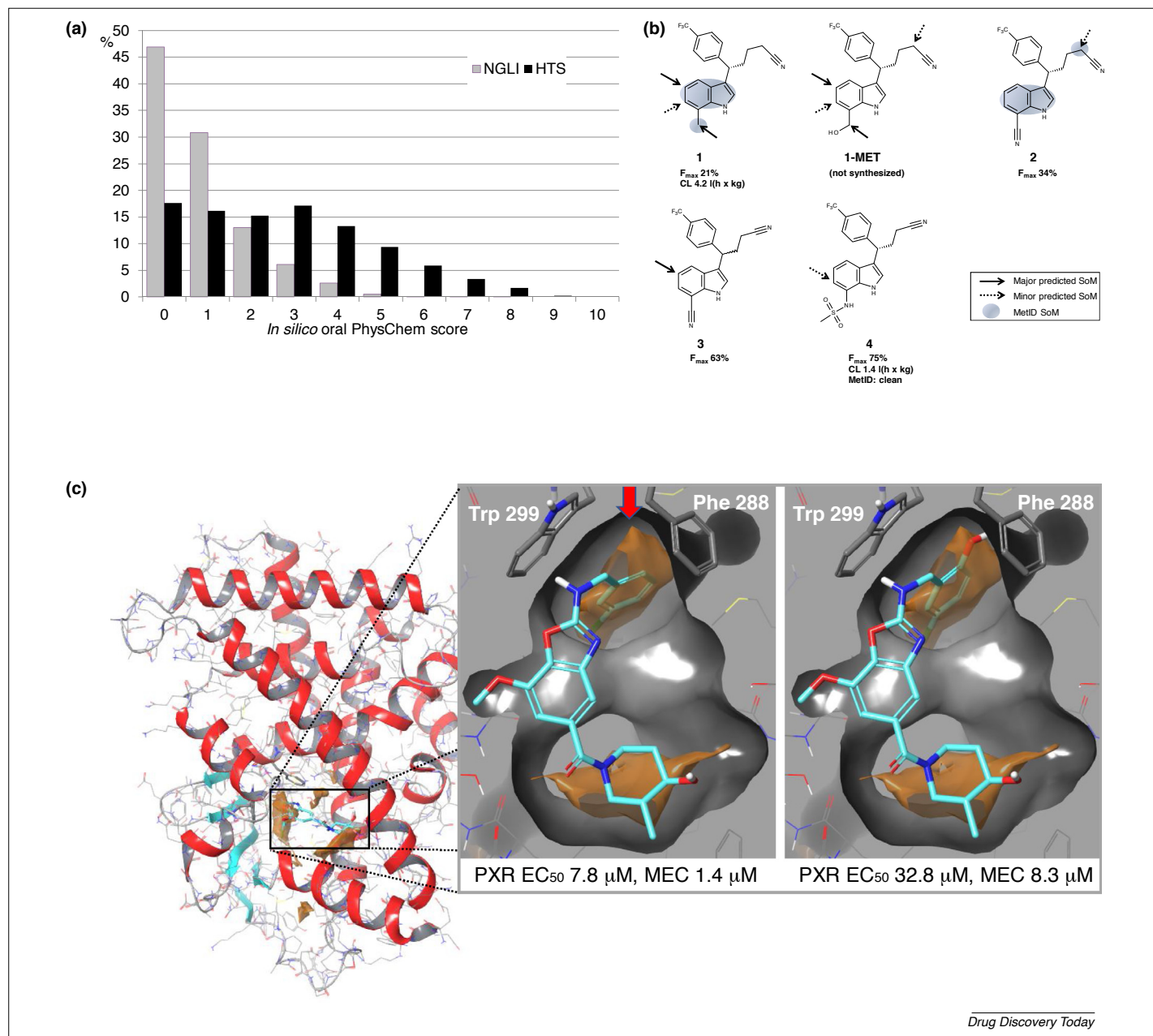


FIGURE 4

Examples of the application of Bayer's absorption, distribution, metabolism, and excretion (ADMET) tools. **(a)** Distribution of oral PhysChem Scores for the compounds from the Next-Generation Library Initiative (NGLI) campaign compared with a historical library used in high-throughput screening (HTS), the oral PhysChem Score being a composite of five predicted physicochemical properties (solubility, topological polar surface area, molecular weight, lipophilicity, and flexibility), the lower the score (can be in the range 0–10) the better; **(b)** representative examples from an mineralocorticoid receptor (MR) antagonist project at Bayer describing the optimization of metabolically labile compounds, and **(c)** the protein structure-based ADMET modeling of reducing pregnane x receptor (PXR) affinity [measured EC_{50} and the minimal effective concentration (MEC) of compounds by chemical modifications].

basicity, which could then be translated into good oral bioavailability in rats [62].

The *in silico* ADMET platform was also an integral part of the Next Generation Library Initiative (NGLI) [9], aiming to enhance the screening collection with 500 000 newly designed compounds, applying Pareto design to achieve favorable physicochemical and ADMET properties [50,63]. Figure 4a shows the distribution of oral PhysChem scores for the NGLI compounds compared with the historic Bayer high-throughput screening (HTS) library, which is significantly left-shifted (favorable compound properties) for the NGLI compounds. The oral PhysChem Score is a composite of five predicted physicochemical properties (solubility, topological polar surface area, molecular weight, lipophilicity, and flexibility). The lower the score (can be in the range 0–10), the better.

Over the years, we developed two methods for regioselectivity modeling of metabolic transformations, namely CypScore [18] and MetScore [17,29], which have been applied in various projects in an attempt to reduce hepatic clearance. One example is a mineralocorticoid receptor antagonists series (Fig. 4b) [64], for which a yet unknown metabolic clearance route was hypothesized.

We have also been looking into protein structure-based ADMET design for several ADMET-relevant proteins [6], but successful examples are rare. Those off-target proteins are characterized by huge, highly flexible ligand-binding sites recognizing a spectrum of ligands. Typically, docking approaches are overburdened in those cases, when X-ray crystallography reveals novel protein conformations with each newly co-crystallized ligand. In an attempt to address this issue, we applied a combined strategy of co-crystallization of novel ligands with pregnane-X-receptor (PXR), overlaying/docking compounds and using the protein structure information to reduce protein–ligand contacts and to overcome persistent Cyp3A4 induction problems [60]. Figure 4c shows a high-affinity thrombin inhibitor [65] with a significant PXR-binding liability. Introduction of a phenolic OH group in a highly lipophilic region in the PXR ligand-binding site (between Trp299 and Phe288, orange surface, red arrow) led to a significant reduction in binding interactions.

Modeling is but one of several tools that can help guide the drug discovery process. The latter remains a challenging process still dominated by trial and error. Modeling has its limitations in terms of how much useful information it can provide, and the development of our modeling program over the past two decades has produced successes as well as failures. Even with our robust internal data set and advanced techniques, modeling ultimately does not eliminate the need for experimental data, especially when it comes to biological experiments.

Concluding remarks and outlook

Over the past two decades, in developing, applying, and experimenting with *in silico* ADMET models, we have learned that the successful application of those tools is mainly dependent on: (i)

the model quality; (ii) model relevance for research processes; and (iii) easy access and interpretability of results. Data, algorithms, and descriptors are all contributing to model quality. A large number of homogenous data and descriptors that are tailored to the underlying experimental endpoint are essential to achieve robust models. The automated generation of large numbers of models (combination of data splits, descriptors, and ML algorithms) and selection of the most accurate ones is helpful. It is also the technical basis for regular model retraining to cover the most current chemical space and to ensure a positive impact on drug discovery projects. The alignment of *in silico* models with readily available *in vitro/in vivo* assays is important for the acceptance and usage of *in silico* approaches in drug discovery projects by many scientists from different fields. Finally, the easy access to *in silico* models is achieved in software tools that allow compound properties to be modeled with a few clicks in seconds. Color coding and estimates of the applicability domain increase the interpretability of the output and ensure usage in drug discovery projects. Although modeling of physicochemical endpoints works reasonably well, the main optimization parameters for oral bioavailability, such as cellular permeation and metabolic clearance, and *in vivo* approaches still need significant improvements. Quantitative models for these endpoints and cytochrome inhibition would be desirable. The limited datasets of single pharmaceutical companies and those that are published might not be enough to reach that goal. Thus, novel approaches of privacy preserving data sharing could be one solution to overcome the lack of data and further advance the field. Additional improvements could come from better 3D-based descriptors of molecules taking intramolecular hydrogen bonds and tautomers into account. Current and future challenges are the proper embedding of *in silico* ADMET models in holistic artificial intelligence approaches (together with estimation of binding affinity and compound synthesizability) and superior solutions for applicability domain estimates.

Conflict of interest

All authors are employees of Bayer AG. A.B., S.S., A.tL., J.W., M.L. and A.H. own stock in Bayer AG.

Acknowledgments

We thank Michael Beck, Matthias Busemann, Djork-Arne Clevert, Ursula Ganzer, Mark Gnoth, Francois Guillou, Michael Grimm, Nikolaus Heinrich, Jörg Keldenich, Ursula Krenz, Dieter Lang, Philip Lienau, Klemens Lustig, Heinrich Meier, Stephan Menz, Britta Nisius, Martin Radke, Andreas Reichel, Karl-Heinz Schlemmer, Rolf Schönnis, Rudolf Schohe-Loop, Thomas Steger-Hartmann, Andreas Sutter, and many other colleagues from Medicinal Chemistry, Toxicology and DMPK for valuable contributions to the *in silico* ADMET platform. Special thanks go to Christian Paulitz-Erdmann and Gabriele Handke-Ergüden for critical reviewing of the manuscript.

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